

Extended Reference List

GAC–MAC 2026 Poster

Building a conceptual 3D geological model of the Wollaston fold-thrust belt, northern Saskatchewan: Implications for pegmatite-related rare earth elements (REE) targeting

Joseph A. Bravo

Department of Geological Sciences, University of Saskatchewan

References

Adeoye, A. (2022). Final report Fanta monazite ore deposit.

Ahmadi, H., & Pekkan, E. (2021). Fault-Based Geological Lineaments Extraction Using Remote Sensing and GIS—A Review. *Geosciences*, 11(5), 183. <https://doi.org/10.3390/geosciences11050183>

Alexandre, P., Kyser, K., Thomas, D., Polito, P., & Marlat, J. (2009). Geochronology of unconformity-related uranium deposits in the Athabasca Basin, Saskatchewan, Canada and their integration in the evolution of the basin. *Mineralium Deposita*, 44(1), 41–59. <https://doi.org/10.1007/s00126-007-0153-3>

Annesley, I. R., Cutford, C., Billard, D., Kusmirski, R. T., Wasyluk, K., Bogdan, T., Sweet, K., & Ludwig, C. (2010). Fraser Lakes Zones A and B, Way Lake Project, Saskatchewan: Geological, geophysical, and geochemical characteristics of basement-hosted uranium mineralization. *Proceedings of the 24th IAGS, Fredericton, 2009. Proceedings of the 24th International Applied Geochemistry Symposium (IAGS)*. https://www.researchgate.net/publication/273765486_Fraser_Lakes_Zones_A_and_B_Way_Lake_Project_Saskatchewan_Geological_geophysical_and_geochemical_characteristics_of_basement-hosted_mineralization

Annesley, I. R., Kamo, S. L., & Kwok, Y. Y. (2009). Major advances from U-Pb geochronology in determining the thermotectonic evolution of the western margin of the Trans-Hudson orogen, eastern Athabasca basement Saskatchewan: Input from Tom Krogh and his co-workers. *GAC–MAC Annual Meeting*.

Annesley, I. R., & Madore, C. (1994). A geological study of the Wollaston–Mudjatik domain boundary in the Wollaston Lake area, Hearne Province, Saskatchewan (R-1230-6-C-94; p. 170). *Saskatchewan Research Council*.

Annesley, I. R., & Madore, C. (1999). Leucogranites and pegmatites from the sub-Athabasca basement, eastern Athabasca Basin, Saskatchewan, Canada: U protore to the unconformity-type deposits.

Annesley, I. R., Madore, C., & Hajnal, Z. (2003). Wollaston-Mudjatik Transition Zone: Its characteristics and influence on the genesis of unconformity-type uranium deposits.

Annesley, I. R., Madore, C., & Portella, P. (2005). Geology and thermotectonic evolution of the western margin of the Trans-Hudson Orogen: Evidence from the eastern sub-Athabasca basement, Saskatchewan. *Canadian Journal of Earth Sciences*, 42(4), 573–597. <https://doi.org/10.1139/e05-034>

Annesley, I. R., McKechnie, C. L., Mercadier, J., Armitage, A., Sexton, A., & Bogdan, T. S. (2013). Fraser Lakes Zone B uranium-thorium-REE deposit, Saskatchewan, Canada: 3D GOCAD modeling to initiate a NI 43-101 inferred resource estimate.

Ansdell, K. M. (2005). Tectonic evolution of the Manitoba-Saskatchewan segment of the Paleoproterozoic Trans-Hudson Orogen, Canada. *Canadian Journal of Earth Sciences*, 42(4), 741–759. <https://doi.org/10.1139/e05-035>

Arienti, G., Bistacchi, A., Caumon, G., Dal Piaz, G., Monopoli, B., & Bertolo, D. (2024). Regional-scale 3D modelling in metamorphic belts: An implicit model-driven workflow applied in the Pennine Alps. *Journal of Structural Geology*, 180, 105045. <https://doi.org/10.1016/j.jsg.2023.105045>

Armitage, A. (2015). Technical report on the Falcon point Uranium project, northern Saskatchewan. GeoVector Management Inc.

Armitage, A., & Sexton, A. (2023). Technical Report on the South Falcon East Property, northern Saskatchewan (p. 101). SGS Geological Services and GeoVector Management Inc.

Astic, T., Heagy, L. J., & Oldenburg, D. W. (2020). Petrophysically and geologically guided multi-physics inversion using a dynamic Gaussian mixture model. *Geophysical Journal International*, 224(1), 40–68. <https://doi.org/10.1093/gji/ggaa378>

Azevedo, L., Grana, D., & Amaro, C. (2019). Geostatistical rock physics AVA inversion. *Geophysical Journal International*, 216(3), 1728–1739. <https://doi.org/10.1093/gji/ggy511>

Balaram, V. (2019). Rare earth elements: A review of applications, occurrence, exploration, analysis, recycling, and environmental impact. *Geoscience Frontiers*, 10(4), 1285–1303. <https://doi.org/10.1016/j.gsf.2018.12.005>

Bauer, T. E., Skyttä, P., Hermansson, T., Dehghannejad, M., & Tavakoli, S. (2015). The Skellefte District. In P. Weihed (Ed.), *3D, 4D and Predictive Modelling of Major Mineral Belts in Europe* (pp. 93–121). Springer International Publishing. https://doi.org/10.1007/978-3-319-17428-0_5

- Beard, C. D., Goodenough, K. M., Borst, A. M., Wall, F., Siegfried, P. R., Deady, E. A., Pohl, C., Hutchison, W., Finch, A. A., Walter, B. F., Elliott, H. A. L., & Brauch, K. (2023). Alkaline-Silicate REE-HFSE Systems. *Economic Geology*, 118(1), 177–208. <https://doi.org/10.5382/econgeo.4956>
- Bickford, M. E., Collerson, K. D., & Lewry, J. F. (1994). Crustal history of the Rae and Hearne provinces, southwestern Canadian Shield, Saskatchewan: Constraints from geochronologic and isotopic data. *Precambrian Research*, 68(1–2), 1–21. [https://doi.org/10.1016/0301-9268\(94\)90062-0](https://doi.org/10.1016/0301-9268(94)90062-0)
- Bickford, M. E., Mock, T. D., Steinhart Iii, W. E., Collerson, K. D., & Lewry, J. F. (2005). Origin of the Archean Sask craton and its extent within the Trans-Hudson orogen: Evidence from Pb and Nd isotopic compositions of basement rocks and post-orogenic intrusions. *Canadian Journal of Earth Sciences*, 42(4), 659–684. <https://doi.org/10.1139/e04-064>
- Bombardieri, D., Duffett, M., McNeill, A., Cracknell, M., & Reading, A. (2021). Insights and Lessons from 3D Geological and Geophysical Modeling of Mineralized Terranes in Tasmania. *Minerals*, 11(11), 1195. <https://doi.org/10.3390/min11111195>
- Bradley, D. C., McCauley, A. D., & Stillings, L. M. (2017). Mineral-deposit model for lithium-cesium-tantalum pegmatites (Nos. 2010–5070–O; Scientific Investigations Report, p. 48). U.S. Geological Survey. <https://doi.org/10.3133/sir20105070O>
- Calcagno, P., Chilès, J. P., Courrioux, G., & Guillen, A. (2008). Geological modelling from field data and geological knowledge I. *Physics of the Earth and Planetary Interiors*, 171(1–4), 147–157. <https://doi.org/10.1016/j.pepi.2008.06.013>
- Cao, X., Liu, Z., Hu, C., Song, X., Quaye, J. A., & Lu, N. (2024). Three-Dimensional Geological Modelling in Earth Science Research: An In-Depth Review and Perspective Analysis. *Minerals*, 14(7), 686. <https://doi.org/10.3390/min14070686>
- Card, C. D., Panã, D., Portella, P., Thomas, D. J., & Annesley, I. R. (2007). Basement rocks to the Athabasca Basin, Saskatchewan and Alberta (No. 588; p. 588). <https://doi.org/10.4095/223745>
- Caumon, G., Collon-Drouaillet, P., Le Carlier de Veslud, C., Viseur, S., & Sausse, J. (2009). Surface-Based 3D Modeling of Geological Structures. *Mathematical Geosciences*, 41(8), 927–945. <https://doi.org/10.1007/s11004-009-9244-2>
- Cloutier, J., Kyser, K., Olivo, G. R., Alexandre, P., & Halaburda, J. (2009). The Millennium Uranium Deposit, Athabasca Basin, Saskatchewan, Canada: An Atypical Basement-Hosted Unconformity-Related Uranium Deposit. *Economic Geology*, 104(6), 815–840. <https://doi.org/10.2113/gsecongeo.104.6.815>
- Corrigan, D., Galley, A., & Pehrsson, S. (2007). Tectonic evolution and metallogeny of the southwestern Trans-Hudson Orogen. In *Mineral Deposits of Canada: A Synthesis of Major*

Deposit-Types, District Metallogeny, the Evolution of Geological Provinces, and Exploration Methods (pp. 881–902). Geological Association of Canada, Mineral Deposits Division.

Corrigan, D., Hajnal, Z., Németh, B., & Lucas, S. B. (2005). Tectonic framework of a Paleoproterozoic arc-continent to continent-continent collisional zone, Trans-Hudson Orogen, from geological and seismic reflection studies. *Canadian Journal of Earth Sciences*, 42(4), 421–434. <https://doi.org/10.1139/e05-025>

Corrigan, D., Pehrsson, S., Wodicka, N., & De Kemp, E. (2009). The Palaeoproterozoic Trans-Hudson Orogen: A prototype of modern accretionary processes. *Geological Society, London, Special Publications*, 327(1), 457–479. <https://doi.org/10.1144/SP327.19>

Cumming, G. L., & Krstic, D. (1992). The age of unconformity-related uranium mineralization in the Athabasca Basin, northern Saskatchewan. *Canadian Journal of Earth Sciences*, 29(8), 1623–1639. <https://doi.org/10.1139/e92-128>

De Kemp, E. A., Schetselaar, E. M., Hillier, M. J., Lydon, J. W., & Ransom, P. W. (2016). Assessing the workflow for regional-scale 3D geologic modeling: An example from the Sullivan time horizon, Purcell Anticlinorium East Kootenay region, southeastern British Columbia. *Interpretation*, 4(3), SM33–SM50. <https://doi.org/10.1190/INT-2015-0191.1>

De La Varga, M., & Wellmann, J. F. (2016). Structural geologic modeling as an inference problem: A Bayesian perspective. *Interpretation*, 4(3), SM1–SM16. <https://doi.org/10.1190/INT-2015-0188.1>

Dentith, M., Enkin, R. J., Morris, W., Adams, C., & Bourne, B. (2020). Petrophysics and mineral exploration: A workflow for data analysis and a new interpretation framework. *Geophysical Prospecting*, 68(1), 178–199. <https://doi.org/10.1111/1365-2478.12882>

Dentith, M., & Mudge, S. T. (2014). *Geophysics for the mineral exploration geoscientist*. Cambridge university press.

Dos Santos, F. P., Leão-Santos, M. H., Borges, W. R., & Borges, P. C. (2024). Geophysical Methods Applied to the Mineralization Discovery of Rare-Earth Elements at the Fazenda Buriti Alkaline Complex, Goiás Alkaline Province, Brazil. *Minerals*, 14(11), 1163. <https://doi.org/10.3390/min14111163>

Dushyantha, N., Batapola, N., Ilankoon, I. M. S. K., Rohitha, S., Premasiri, R., Abeysinghe, B., Ratnayake, N., & Dissanayake, K. (2020). The story of rare earth elements (REEs): Occurrences, global distribution, genesis, geology, mineralogy and global production. *Ore Geology Reviews*, 122, 103521. <https://doi.org/10.1016/j.oregeorev.2020.103521>

Enkin, R. J., Hamilton, T. S., & Morris, W. A. (2020). The Henkel Petrophysical Plot: Mineralogy and Lithology From Physical Properties. *Geochemistry, Geophysics, Geosystems*, 21(1), e2019GC008818. <https://doi.org/10.1029/2019GC008818>

Fiolleau, T. (2023). Technical Report (2023) Kulyk Lake Property | Saskatchewan, Canada [NI 43-101]. Axiom Exploraiton GROUP LTD.

Forster, R. W. (1970). A geological report on dynamic petroleum products ltd. In Permit (Issue 1, pp. 1–36).

Garcia, X., & Jones, A. G. (2005). Electromagnetic image of the Trans-Hudson orogen THO94 transect. *Canadian Journal of Earth Sciences*, 42(4), 479–493. <https://doi.org/10.1139/e05-016>

Global Business Reports. (2024). Brazil Mining. Global Business Reports.

Government of Canada. (2022, April 7). Critical minerals: An opportunity for Canada [Campaigns]. Canada.Ca. <https://www.canada.ca/en/campaign/critical-minerals-in-canada/critical-minerals-an-opportunity-for-canada.html>

Grose, L., Ailleres, L., Laurent, G., & Jessell, M. (2021). LoopStructural 1.0: Time-aware geological modelling. *Geoscientific Model Development*, 14(6), 3915–3937. <https://doi.org/10.5194/gmd-14-3915-2021>

Groshong, R. H. (2006). 3-D structural geology: A practical guide to surface and subsurface map interpretation (2nd ed). Springer.

Guffey, S., Piercey, S., Ansdell, K., Kyser, K., Kotzer, T., Quirt, D., & Zaluski, G. (2019). Geochemical footprint of the Millennium unconformity-type uranium deposit, Canada: Implications for vectoring new targets. *Geochemistry: Exploration, Environment, Analysis*, 19(4), 395–413. <https://doi.org/10.1144/geochem2018-036>

Hajnal, Z., Annesley, I. R., White, D., Matthews, R. B., Sopuck, V., Koch, R., Leppin, M., & Ahuja, S. (1997). Sedimentary-Hosted Mineral Deposits: A High-Resolution Seismic Survey in the Athabasca Basin. *Exploration '97: Geophysics and Geochemistry at the Millennium. Fourth Decennial International Conference on Mineral Exploration*. <https://www.dmec.ca/publications/exploration-97/>

Hajnal, Z., Lewry, J., White, D., Ashton, K., Clowes, R., Stauffer, M., Gyorfi, I., & Takacs, E. (2005). The Sask Craton and Hearne Province margin: Seismic reflection studies in the western Trans-Hudson Orogen. *Canadian Journal of Earth Sciences*, 42(4), 403–419. <https://doi.org/10.1139/e05-026>

Hajnal, Z., White, D. J., Takacs, E., Gyorfi, I., Annesley, I. R., Wood, G., O'Dowd, C., & Nimeck, G. (2010). Application of modern 2-D and 3-D seismic-reflection techniques for uranium exploration in the Athabasca Basin This article is one of a series of papers published in this Special Issue on the theme Lithoprobe—Parameters, processes, and the evolution of a continent. *Canadian Journal of Earth Sciences*, 47(5), 761–782. <https://doi.org/10.1139/E10-026>

- Hasoon, S. K., & Farman, G. M. (2024). A Comprehensive Review for Integrating Petrophysical Properties, Rock Typing, and Geological Modeling for Enhanced Reservoir Characterization. *Journal of Engineering*, 30(10), 69–101. <https://doi.org/10.31026/j.eng.2024.10.05>
- Hoffman, P. F. (1988). United Plates of America, The Birth of a Craton: Early Proterozoic Assembly and Growth of Laurentia. *Annual Review of Earth and Planetary Sciences*, 16(1), 543–603. <https://doi.org/10.1146/annurev.ea.16.050188.002551>
- Hoffman, P. F. (1989). Precambrian geology and tectonic history of North America. In A. W. Bally & A. R. Palmer (Eds.), *The Geology of North America—An Overview* (pp. 447–512). Geological Society of America. <https://doi.org/10.1130/DNAG-GNA-A.447>
- Jeanneret, P., Goncalves, P., Durand, C., Poujol, M., Trap, P., Marquer, D., Quirt, D., & Ledru, P. (2017). Geochronological constraints on the trans-Hudsonian tectono-metamorphic evolution of the pre-Athabasca basement within the Wollaston-Mudjatik Transition Zone, Saskatchewan. *Precambrian Research*, 301, 152–178. <https://doi.org/10.1016/j.precamres.2017.07.019>
- Jeanneret, P., Goncalves, P., Durand, C., Trap, P., Marquer, D., Quirt, D., & Ledru, P. (2016). Tectono-metamorphic evolution of the pre-Athabasca basement within the Wollaston–Mudjatik Transition Zone, Saskatchewan. *Canadian Journal of Earth Sciences*, 53(3), 231–259. <https://doi.org/10.1139/cjes-2015-0136>
- Jefferson, C. W., Thomas, D. J., Gandhi, S. S., Ramaekers, P., Delaney, G., Brisbin, D., Cutts, C., Portella, P., & Olson, R. A. (2007). Unconformity-associated uranium deposits of the Athabasca Basin, Saskatchewan and Alberta. 588. <https://doi.org/10.4095/223744>
- Jessell, M. (2001). Three-dimensional geological modelling of potential-field data. *Computers & Geosciences*, 27(4), 455–465. [https://doi.org/10.1016/S0098-3004\(00\)00142-4](https://doi.org/10.1016/S0098-3004(00)00142-4)
- Jessell, M., Aillères, L., Kemp, E. D., Lindsay, M., Wellmann, F., Hillier, M., Laurent, G., Carmichael, T., & Martin, R. (2014). Next Generation Three-Dimensional Geologic Modeling and Inversion. In K. D. Kelley & H. C. Golden, *Building Exploration Capability for the 21st Century* (p. 12). Society of Economic Geologists. <https://doi.org/10.5382/SP.18.13>
- Jessell, M. W., Ailleres, L., & De Kemp, E. A. (2010). Towards an integrated inversion of geoscientific data: What price of geology? *Tectonophysics*, 490(3–4), 294–306. <https://doi.org/10.1016/j.tecto.2010.05.020>
- Jones, A. G., Ledo, J., & Ferguson, I. J. (2005). Electromagnetic images of the Trans-Hudson orogen: The North American Central Plains anomaly revealed. *Canadian Journal of Earth Sciences*, 42(4), 457–478. <https://doi.org/10.1139/e05-018>
- Jordens, A., Cheng, Y. P., & Waters, K. E. (2013). A review of the beneficiation of rare earth element bearing minerals. *Minerals Engineering*, 41, 97–114. <https://doi.org/10.1016/j.mineng.2012.10.017>

Ko, C. B. (1971). Geological report on dynamic petroleum products ltd. CBS 1837, sask (pp. 1–23). N.T.S.

Lajaunie, C., Courrioux, G., & Manuel, L. (1997). Foliation fields and 3D cartography in geology: Principles of a method based on potential interpolation. *Mathematical Geology*, 29(4), 571–584. <https://doi.org/10.1007/BF02775087>

Lelièvre, P. G., & Oldenburg, D. W. (2009). A comprehensive study of including structural orientation information in geophysical inversions. *Geophysical Journal International*, 178(2), 623–637. <https://doi.org/10.1111/j.1365-246X.2009.04188.x>

Lesage, G., Byrne, K., Morris, W. A., Enkin, R. J., Lee, R. G., Mir, R., & Hart, C. J. R. (2019). Interpreting regional 3D fault networks from integrated geological and geophysical data sets: An example from the Guichon Creek batholith, British Columbia. *Journal of Structural Geology*, 119, 93–106. <https://doi.org/10.1016/j.jsg.2018.12.007>

Lewry, J. F., & Sibbald, T. I. I. (1980). Thermotectonic evolution of the Churchill Province in Northern Saskatchewan. *Tectonophysics*, 68(1–2), 45–82. [https://doi.org/10.1016/0040-1951\(80\)90008-6](https://doi.org/10.1016/0040-1951(80)90008-6)

Li, Z., Chi, G., & Bethune, K. M. (2016). The effects of basement faults on thermal convection and implications for the formation of unconformity-related uranium deposits in the Athabasca Basin, Canada. *Geofluids*, 16(4), 729–751. <https://doi.org/10.1111/gfl.12180>

Li, Z., Chi, G., Bethune, K. M., Thomas, D., & Zaluski, G. (2017). Structural Controls on Fluid Flow During Compressional Reactivation of Basement Faults: Insights from Numerical Modeling for the Formation of Unconformity-Related Uranium Deposits in the Athabasca Basin, Canada. *Economic Geology*, 112(2), 451–466. <https://doi.org/10.2113/econgeo.112.2.451>

Lindsay, M. D., Aillères, L., Jessell, M. W., De Kemp, E. A., & Betts, P. G. (2012). Locating and quantifying geological uncertainty in three-dimensional models: Analysis of the Gippsland Basin, southeastern Australia. *Tectonophysics*, 546–547, 10–27. <https://doi.org/10.1016/j.tecto.2012.04.007>

Lindsay, M. D., Jessell, M. W., Aillères, L., Perrouy, S., De Kemp, E., & Betts, P. G. (2013). Geodiversity: Exploration of 3D geological model space. *Tectonophysics*, 594, 27–37. <https://doi.org/10.1016/j.tecto.2013.03.013>

Lucas, S. B., Green, A., Hajnal, Z., White, D., Lewry, J., Ashton, K., Weber, W., & Clowes, R. (1993). Deep seismic profile across a Proterozoic collision zone: Surprises at depth. *Nature*, 363(6427), 339–342. <https://doi.org/10.1038/363339a0>

Lucas, S. B., White, D., Hajnal, Z., Lewry, J., Green, A., Clowes, R., Zwanzig, H., Ashton, K., Schledewitz, D., Stauffer, M., Norman, A., Williams, P. F., & Spence, G. (1994). Three-

dimensional collisional structure of the Trans-Hudson Orogen, Canada. *Tectonophysics*, 232(1–4), 161–178. [https://doi.org/10.1016/0040-1951\(94\)90082-5](https://doi.org/10.1016/0040-1951(94)90082-5)

Magee, T., & Cubbon, C. (2022). Kuylk Prospecting Program_Phase 1 [LOGISTICS & PROJECT SUMMARY REPORT].

Mancheri, N. A., Sprecher, B., Bailey, G., Ge, J., & Tukker, A. (2019). Effect of Chinese policies on rare earth supply chain resilience. *Resources, Conservation and Recycling*, 142, 101–112. <https://doi.org/10.1016/j.resconrec.2018.11.017>

Martinez, C., Mulè, S., & Blaikie, T. (2021). Enhancing interpretation of geophysical models using petrophysical logging.

Mawdsley, B. (1957). The Geology of the Middle Foster Lakee Area Northern Saskatchewan.

Maxeiner, R. O., Rayner, N. M., Ansdell, K. M., & Creaser, R. A. (2022). Constraints on the evolution of the eastern margin of the Hearne craton: New data from the southwestern Rottenstone Domain, Trans-Hudson Orogen, Saskatchewan. *Precambrian Research*, 381, 106851. <https://doi.org/10.1016/j.precamres.2022.106851>

McCafferty, A. E., Phillips, J. D., & Driscoll, R. L. (2016). Magnetic and Gravity Gradiometry Framework for Mesoproterozoic Iron Oxide-Apatite and Iron Oxide-Copper-Gold Deposits, Southeast Missouri. *Economic Geology*, 111(8), 1859–1882. <https://doi.org/10.2113/econgeo.111.8.1859>

McCafferty, A. E., Phillips, J. D., Hofstra, A. H., & Day, W. C. (2019). Crustal architecture beneath the southern Midcontinent (USA) and controls on Mesoproterozoic iron-oxide mineralization from 3D geophysical models. *Ore Geology Reviews*, 111, 102966. <https://doi.org/10.1016/j.oregeorev.2019.102966>

McFarlane, C. R. M., & McKeough, M. (2013). Petrogenesis of the Kulyk Lake monazite-apatite-Fe(Ti)-oxide occurrence revealed using in-situ LA-(MC)-ICP-MS trace element mapping, U-Pb dating, and Sm-Nd isotope systematics on monazite. *American Mineralogist*, 98(10), 1644–1659. <https://doi.org/10.2138/am.2013.4368>

McKechnie, C. L. (2012). The Fraser Lakes Zone B U-Th-REE deposit and its host rocks: Implications for pegmatite- and leucogranite-hosted U-Th-REE Deposits in northern Saskatchewan, Canada [Master's thesis, University of Saskatchewan]. <http://hdl.handle.net/10388/ETD-2012-12-879>

McKechnie, C. L., Annesley, I. R., & Ansdell, K. M. (2012a). Medium- to low-pressure pelitic gneisses of Fraser Lakes Zone B, Wollaston Domain, northern Saskatchewan, Canada: Mineral compositions, metamorphic P-T-t path, and implications for the genesis of radioactive abyssal granitic pegmatites. *The Canadian Mineralogist*, 50(6), 1669–1694. <https://doi.org/10.3749/canmin.50.6.1669>

- McKechnie, C. L., Annesley, I. R., & Ansdell, K. M. (2012b). Radioactive abyssal granitic pegmatites and leucogranites in the Wollaston Domain, northern Saskatchewan, Canada: Mineral compositions and conditions of emplacement in the Fraser Lakes area. *The Canadian Mineralogist*, 50(6), 1637–1667. <https://doi.org/10.3749/canmin.50.6.1637>
- McKechnie, C. L., Annesley, I. R., & Ansdell, K. M. (2013). Geological Setting, Petrology, and Geochemistry of Granitic Pegmatites and Leucogranites Hosting U-Th-REE Mineralization at Fraser Lakes Zone B, Wollaston Domain, Northern Saskatchewan, Canada.
- McKeough, M. (2013). The Geochemical, Geochronologic, and Isotopic Constraints on U-Th-REE Pegmatites and related mineralization of south-central part of the the Wollaston Domain, Northern Saskatchewan, Canada [Master's thesis]. University of New Brunswick.
- McKeough, M. A., & Lentz, D. R. (2011). Paleoproterozoic Late-tectonic Granitic Pegmatite-hosted U-Th±REE-Y-Nb Mineralization, Northern Saskatchewan: Products of Assimilation, Fractional Crystallization, and Hybridization Processes.
- McKeough, M. A., Lentz, D. R., & Brown, J. (2010). Geology and Associated Pegmatite- and Vein-hosted Uranium Mineralization of the Kulyk, Eagle, and Karin Lakes Regions, Wollaston Domain, Northern Saskatchewan.
- McKeough, M. A., Lentz, D. R., McFarlane, C. R. M., & Brown, J. (2013). Geology and evolution of pegmatite-hosted U-Th ± REE-Y-Nb Mineralization, Kulyk, Eagle, and Karin Lakes region, Wollaston Domain, northern Saskatchewan, Canada: Examples of the dual role of extreme fractionation and hybridization processes. *Journal of GEOsciences*, 321–346. <https://doi.org/10.3190/jgeosci.153>
- Mckinsey & Company. (2024). Global materials perspective 2024 [Industry Report]. Mckinsey & Company. <https://www.mckinsey.com/industries/energy-and-materials/our-insights/global-materials-perspective>
- Mercadier, J., Annesley, I. R., McKechnie, C. L., Bogdan, T. S., & Creighton, S. (2013). Magmatic and Metamorphic Uraninite Mineralization in the Western Margin of the Trans-Hudson Orogen (Saskatchewan, Canada): A Uranium Source for Unconformity-Related Uranium Deposits? *Economic Geology*, 108(5), 1037–1065. <https://doi.org/10.2113/econgeo.108.5.1037>
- Mints, M. V., Dokukina, K. A., & Afonina, T. B. (2021). Deep crustal structure and Palaeoproterozoic evolution of the supercontinent Lauroscandia: 3D model of Trans-Hudson Orogen and new insight based on LITHOPROBE, FIRE and 1-EU reflection seismic and HuBLE tomography imaging. *Tectonophysics*, 820, 229119. <https://doi.org/10.1016/j.tecto.2021.229119>
- Montsion, R. M., Perrouy, S., Lindsay, M. D., Jessell, M. W., & Frieman, B. M. (2021). Mapping structural complexity using geophysics: A new geostatistical approach applied to greenstone belts of the southern Superior Province, Canada. *Tectonophysics*, 812, 228889. <https://doi.org/10.1016/j.tecto.2021.228889>

Müller, A., Brönnner, M., Menuge, J., Williamson, B., Haase, C., Tassis, G., Pohl, C., Brauch, K., Saalmann, K., Teodoro, A., Roda-Robles, E., Cardoso-Fernandes, J., Smith, K., Wall, F., Lima, A., Santos, D., Hopfner, M., Garate-Olave, I., Errandonea-Martin, J., ... Lloret, C. P. (2025). The GREENPEG Project Toolset to Explore for Buried Pegmatites Hosting Lithium, High-Purity Quartz, and Other Critical Raw Materials.

Müller, A., Reimer, W., Wall, F., Williamson, B., Menuge, J., Brönnner, M., Haase, C., Brauch, K., Pohl, C., Lima, A., Teodoro, A., Cardoso-Fernandes, J., Roda-Robles, E., Harrop, J., Smith, K., Wanke, D., Unterweissacher, T., Hopfner, M., Schröder, M., ... Rausa, A. (2023). GREENPEG – exploration for pegmatite minerals to feed the energy transition: First steps towards the Green Stone Age. *Geological Society, London, Special Publications*, 526(1), 193–218. <https://doi.org/10.1144/SP526-2021-189>

Németh, B., Clowes, R. M., & Hajnal, Z. (2005). Lithospheric structure of the Trans-Hudson Orogen from seismic refraction—Wide-angle reflection studies. *Canadian Journal of Earth Sciences*, 42(4), 435–456. <https://doi.org/10.1139/e05-032>

Normand, C. (2014). Report 264: Rare Earths in Saskatchewan: Mineralization Types, Settings, and Distributions. <https://pubsaskdev.blob.core.windows.net/pubsask-prod/91564/91564-English.pdf>

Partridge, E. F. (1968). Prospecting notes, Alexander-Fraser Lakes area. Unpublished field notes (p. 74 02-0002).

Pinto-Ward, C. (2018). Controls on the Enrichment of the Serra Verde Rare Earth Deposit, Brazil [PhD dissertation, Imperial College London]. <https://spiral.imperial.ac.uk/entities/publication/cc48f550-5bde-42e3-969b-d41d1bc47a36>

Pirot, G., Joshi, R., Giraud, J., Lindsay, M. D., & Jessell, M. W. (2022). loopUI-0.1: Indicators to support needs and practices in 3D geological modelling uncertainty quantification. *Geoscientific Model Development*, 15(12), 4689–4708. <https://doi.org/10.5194/gmd-15-4689-2022>

Poliakovska, K. (2024). Integrated geological-geophysical analysis of REE mineralization from two selected areas in the Canadian and Ukrainian Precambrian Shields [PhD dissertation]. Université de Lorraine.

Poliakovska, K., Annesley, I. R., & Hajnal, Z. (2023). Geophysical Constraints to the Geological Evolution and Genesis of Rare Earth Element–Thorium–Uranium Mineralization in Pegmatites at Alces Lake, SK, Canada. *Minerals*, 14(1), 25. <https://doi.org/10.3390/min14010025>

Poliakovska, K., Annesley, I. R., Ivanik, O., Sykes, J., Guest, N., & Otsuki, A. (2021). Geological setting, petrology, and 3D geochemical modeling of REE-Th-U mineralization in the Alces Lake area, Northern Saskatchewan, Canada.

Portella, P., & Annesley, I. R. (2000). Paleoproterozoic Thermotectonic Evolution of the Eastern Sub-Athabasca Basement, Northern Saskatchewan: Integrated Geophysical and Geological Data. Saskatchewan Industry and Resources, Geological Survey, Miscellaneous Report, 2, 1–15.

Ramaekers, P., Jefferson, C. W., Yeo, G. M., Collier, B., Long, D. G. F., Drever, G., McHardy, S., Jiricka, D., Cutts, C., Wheatley, K., Catuneanu, O., Bernier, S., Kupsch, B., & Post, R. (2007). Revised geological map and stratigraphy of the Athabasca Group, Saskatchewan and Alberta (No. 588; p. 588). <https://doi.org/10.4095/223754>

Ray, G. E. (1977). Compilation Geology, Foster Lake Area (74A), including Reconnaissance Geological Mapping of the Daly Lake (West) part of the Middle Foster Lake Areas. Saskatchewan Geological Survey. <http://publications.gov.sk.ca>

Ray, G. E. (1981). Bedrock Geology and Geochemistry, Daly Lake (West) Area and Part of Middle Foster Lake Area (No. 208). Saskatchewan Geological Survey. <http://publications.gov.sk.ca>

Ray, G. E., & Wanless, R. K. (1980). The age and geological history of the Wollaston, Peter Lake, and Rottenstone domains in northern Saskatchewan. Canadian Journal of Earth Sciences, 17(3), 333–347. <https://doi.org/10.1139/e80-033>

Rayner, N. M., Stern, R. A., & Bickford, M. E. (2005). Tectonic implications of new SHRIMP and TIMS UPb geochronology of rocks from the Sask Craton, Peter Lake Domain, and Hearne margin, Trans-Hudson Orogen, Saskatchewan. Canadian Journal of Earth Sciences, 42(4), 635–657. <https://doi.org/10.1139/e04-045>

Roering, J. J., Mackey, B. H., Marshall, J. A., Sweeney, K. E., Deligne, N. I., Booth, A. M., Handwerker, A. L., & Cerovski-Darriau, C. (2013). ‘You are HERE’: Connecting the dots with airborne lidar for geomorphic fieldwork. Geomorphology, 200, 172–183. <https://doi.org/10.1016/j.geomorph.2013.04.009>

Rogers, S. L., Giles, S., Dowey, N., Greene, S. E., Bhatia, R., Van Landeghem, K., & King, C. (2024). “you just look at rocks, and have beards” Perceptions of Geology From the United Kingdom: A Qualitative Analysis From an Online Survey. Earth Science, Systems and Society, 4(1), 10078. <https://doi.org/10.3389/esss.2024.10078>

Sameera, K. A. G., Wickramasinghe, W. A. G. K., Harankahawa, S. B., Welikanna, C. R., & De Silva, K. T. U. S. (2020). Radiometric surveying for Th and U mineralization in southwestern, Sri Lanka: Radiological, mineralogical and geochemical characteristics of the radioactive anomalies. Journal of the Geological Society of Sri Lanka, 21(2), 57. <https://doi.org/10.4038/jgssl.v21i2.49>

Santana, I. V., & Botelho, N. F. (2022). REE residence, behaviour and recovery from a weathering profile related to the Serra Dourada Granite, Goiás/Tocantins States, Brazil. Ore Geology Reviews, 143, 104751. <https://doi.org/10.1016/j.oregeorev.2022.104751>

Schetselaar, E., Ames, D., & Grunsky, E. (2017). Integrated 3D Geological Modeling to Gain Insight in the Effects of Hydrothermal Alteration on Post-Ore Deformation Style and Strain Localization in the Flin Flon Volcanogenic Massive Sulfide Ore System. *Minerals*, 8(1), 3. <https://doi.org/10.3390/min8010003>

Schetselaar, E. M. (2013). Mapping the 3D lithofacies architecture of a VMS ore system on a curvilinear-faulted grid: A case study from the Flin Flon mining camp, Canada. *Ore Geology Reviews*, 53, 261–275. <https://doi.org/10.1016/j.oregeorev.2013.01.012>

Scibek, J., & Annesley, I. R. (2021). Permeability Testing of Drill Core from Basement Rocks in the Fault-Hosted Gryphon U Deposit (Eastern Athabasca Basin, Canada): Insights into Fluid–Rock Interactions Related to Deposit Formation and Redistribution. *Natural Resources Research*, 30(4), 2909–2956. <https://doi.org/10.1007/s11053-021-09811-x>

Shah, A. K., Taylor, R. D., Walsh, G. J., & Phillips, J. D. (2021). Integrated geophysical imaging of rare earth element-bearing iron oxide-apatite deposits in the Eastern Adirondack Highlands, New York. *GEOPHYSICS*, 86(1), B37–B54. <https://doi.org/10.1190/geo2019-0783.1>

Sinding-Larsen, R., & Wellmer, F.-W. (Eds.). (2012). *Non-Renewable Resource Issues: Geoscientific and Societal Challenges*. Springer Netherlands. <https://doi.org/10.1007/978-90-481-8679-2>

Staněk, F., Franěk, J., Jelínek, J., & Žáček, V. (2025). Estimating relative uncertainty of geological 3D models with low density of input data in geologically complex regions. *Earth Science Informatics*, 18(2), 259. <https://doi.org/10.1007/s12145-025-01778-0>

Stoch, B., Anthonissen, C. J., McCall, M.-J., Basson, I. J., Deacon, J., Cloete, E., Botha, J., Britz, J., Strydom, M., Nel, D., & Bester, M. (2018). 3D implicit modeling of the Sishen Mine: New resolution of the geometry and origin of Fe mineralization. *Mineralium Deposita*, 53(6), 835–853. <https://doi.org/10.1007/s00126-017-0784-y>

Sweetapple, M. T., Grigson, M. W., Tornatora, P., & Urgine, S. (2019). The Archean Mt. Cattlin spodumene pegmatite group and 3D geochemical mapping of large “unzoned” pegmatites of economic significance. *The Canadian Mineralogist*, 57(5), 803–805. <https://doi.org/10.3749/canmin.AB00026>

Takacs, E., Hajnal, Z., Pandit, B., & Annesley, I. R. (2015). Mapping of alteration zones with seismic-amplitude data and well logs in the hard-rock environment of the Keefe Lake area, Athabasca Basin, Canada. *The Leading Edge*, 34(5), 530–538. <https://doi.org/10.1190/tle34050530.1>

Tavakoli, S., Sarlus, Z., Kronsell, I., & Bauer, T. E. (2021). 2.5D geophysical model of the Gällivare mining area: An integrated study to model the top 4 km of the subsurface and guide for future exploration activities. *Geophysical Prospecting*, 69(4), 821–841. <https://doi.org/10.1111/1365-2478.13077>

- Thomas, M. D., Ford, K. L., & Keating, P. (2016). Review paper: Exploration geophysics for intrusion-hosted rare metals. *Geophysical Prospecting*, 64(5), 1275–1304. <https://doi.org/10.1111/1365-2478.12352>
- Toma, J., Creaser, R. A., Card, C., Pana, D., Dufrane, A., & Li, L. (2024). Nuna supercontinent assembly linked to carbon cycling in shear zones 1.9–1.7 billion years ago. *Nature Geoscience*, 17(10), 1038–1045. <https://doi.org/10.1038/s41561-024-01519-w>
- Torvela, T., Moreau, J., Butler, R. W. H., Korja, A., & Heikkinen, P. (2013). The mode of deformation in the orogenic mid-crust revealed by seismic attribute analysis. *Geochemistry, Geophysics, Geosystems*, 14(4), 1069–1086. <https://doi.org/10.1002/ggge.20050>
- Tran, H., Ansdell, K., Bethune, K., Ashton, K., & Hamilton, M. (2008). Provenance and tectonic setting of Paleoproterozoic metasedimentary rocks along the eastern margin of Hearne craton: Constraints from SHRIMP geochronology, Wollaston Group, Saskatchewan, Canada. *Precambrian Research*, 167(1–2), 171–185. <https://doi.org/10.1016/j.precamres.2008.08.003>
- Tran, H. T., Ansdell, K., Bethune, K., Watters, B., & Ashton, K. (2003). Nd isotope and geochemical constraints on the depositional setting of Paleoproterozoic metasedimentary rocks along the margin of the Archean Hearne craton, Saskatchewan, Canada. *Precambrian Research*, 123(1), 1–28. [https://doi.org/10.1016/S0301-9268\(03\)00012-3](https://doi.org/10.1016/S0301-9268(03)00012-3)
- Tran, H. T., Yeo, G., & Bradley, S. (1998). Geology of the Daly-Suttle-Middle Foster Lakes Area, Eastern Wollaston Domain (NTS 74A-5, -11, and -12).
- Unger, G., Zimmermann, R., & Gloaguen, R. (2018). 3D Modeling of the Epembe (Namibia) Nb-Ta-P-(LREE) Carbonatite Deposit: New Insights into Geometry Related to Rare Metal Enrichment. *Minerals*, 8(12), 600. <https://doi.org/10.3390/min8120600>
- Veress, E., Andersson, J. B. H., Popova, I., Annesley, I. R., & Bauer, T. E. (2024). Three-Dimensional Geologic Modeling of the Kiruna Mining District, Sweden: Insights into the Crustal Architecture and Structural Controls on Iron Oxide-Apatite Mineralization. *Economic Geology*, 119(5), 1089–1113. <https://doi.org/10.5382/econgeo.5080>
- Veress, E., Rydman, O., Farrenzo, R. A., Andersson, J. B. H., Warlo, M., Annesley, I. R., & Bauer, T. E. (2025). Linking mineralogical, geochemical, and textural controls to petrophysical signatures in the Kiruna Mining District, Sweden. *Ore and Energy Resource Geology*, 19, 100114. <https://doi.org/10.1016/j.oreoa.2025.100114>
- Wang, Z.-Y., Fan, H.-R., Zhou, L., Yang, K.-F., & She, H.-D. (2020). Carbonatite-Related REE Deposits: An Overview. *Minerals*, 10(11), 965. <https://doi.org/10.3390/min10110965>
- Watkinson, D. H., & Mainwaring, P. R. (1976). The Kulyk Lake monazite deposit, northern Saskatchewan. *Canadian Journal of Earth Sciences*, 13(3), 470–475. <https://doi.org/10.1139/e76-048>

Weihed, P. (Ed.). (2015). 3D, 4D and Predictive Modelling of Major Mineral Belts in Europe. Springer International Publishing. <https://doi.org/10.1007/978-3-319-17428-0>

Wellmann, F., & Caumon, G. (2018). 3-D Structural geological models: Concepts, methods, and uncertainties. In *Advances in Geophysics* (Vol. 59, pp. 1–121). Elsevier. <https://doi.org/10.1016/bs.agph.2018.09.001>

Wellmann, J. F., Horowitz, F. G., Schill, E., & Regenauer-Lieb, K. (2010). Towards incorporating uncertainty of structural data in 3D geological inversion. *Tectonophysics*, 490(3–4), 141–151. <https://doi.org/10.1016/j.tecto.2010.04.022>

Wellmann, J. F., & Regenauer-Lieb, K. (2012). Uncertainties have a meaning: Information entropy as a quality measure for 3-D geological models. *Tectonophysics*, 526–529, 207–216. <https://doi.org/10.1016/j.tecto.2011.05.001>

White, D. J., Hajnal, Z., Roberts, B., Györfi, I., Reilkoff, B., Bellefleur, G., Mueller, C., Woelz, S., Mwenifumbo, C. J., Takács, E., Schmitt, D. R., Brisbin, D., Jefferson, C. W., Koch, R., Powell, B., & Annesley, I. R. (2007). Seismic methods for uranium exploration: An overview of EXTECH IV seismic studies at the McArthur River mining camp, Athabasca Basin, Saskatchewan (No. 588; p. 588). <https://doi.org/10.4095/223782>

Whitmeyer, S. J., & Karlstrom, K. E. (2007). Tectonic model for the Proterozoic growth of North America. <https://doi.org/10.1130/GES00055.1>

Witter, J. B., & Melosh, G. (2018). The Value and Limitations of 3D Models for Geothermal Exploration.

Yeh, C.-H., Lu, Y.-C., Khoshnevisan, S., Juang, C. H., Tien, Y.-M., & Dong, J.-J. (2024). LiDAR-based 3D litho-stratigraphic models calibrated with limited boreholes. *Engineering Geology*, 332. <https://doi.org/10.1016/j.enggeo.2024.107461>

Yeo, G. M., & Delaney, G. (2007). The Wollaston Supergroup, stratigraphy and metallogeny of a Paleoproterozoic Wilson cycle in the Trans-Hudson Orogen, Saskatchewan (No. 588; p. 588). <https://doi.org/10.4095/223746>

Yu, Y., Li, Y., Liu, Y., Ling, X.-X., Wu, L.-G., Yang, L., Yang, L., Yang, B., Zhao, Y.-G., & Li, X.-H. (2024). Three-stage niobium mineralization at Bayan Obo, China. *National Science Review*, 11(4), 063. <https://doi.org/10.1093/nsr/nwae063>